



University of Amsterdam
Recommender Systems

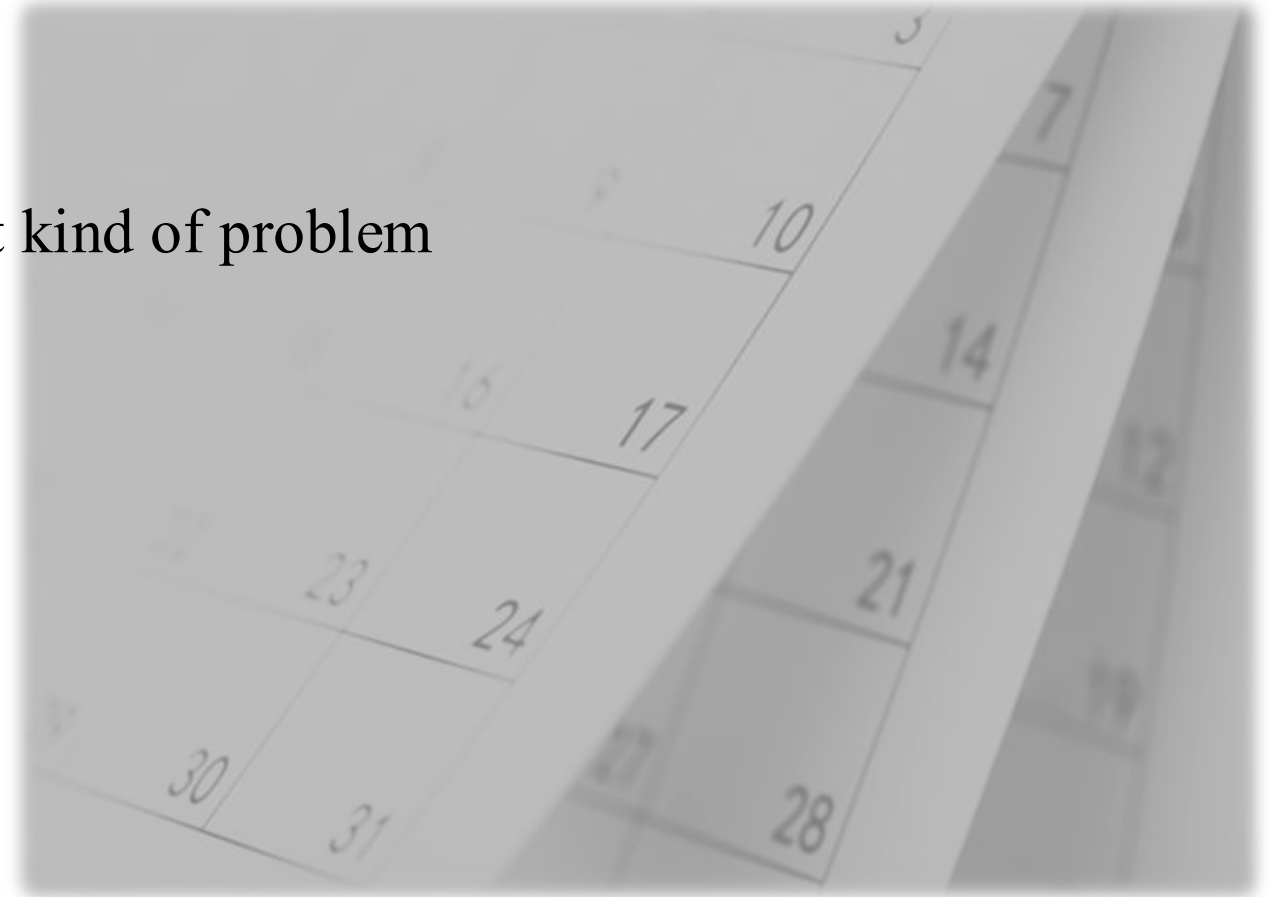
News Recommender Systems

Dr. Dina Strikovic | Meeting 3



Contents

- Recap: Data Bias
 - Different kind of bias = different kind of problem
- News Environment
 - Then and now
 - What are the implications?
 - What's the future?
- An activity in there somewhere





Recap: Bias

Three kinds of bias (that we focused on)

- Data Bias: insufficient data; over-/under-representation of sample; etc.
- **System Bias: popularity and exposure biases**
- Human Bias: human vs. machine tradeoffs

News Recommender
Algorithms



News media: what is it good for?

News media: what is it good for?



Gatekeeping

- Process through which information is filtered before it's disseminated

→ Who are the gatekeepers?

→ What criteria do they use to filter information?

News media: what is it good for?



Gatekeeping

- In the past: journalists and editors select news
- Perspective of the news organization that defines criteria

→ Any problems with this?

News media: what is it good for?





News media: what is it good for?



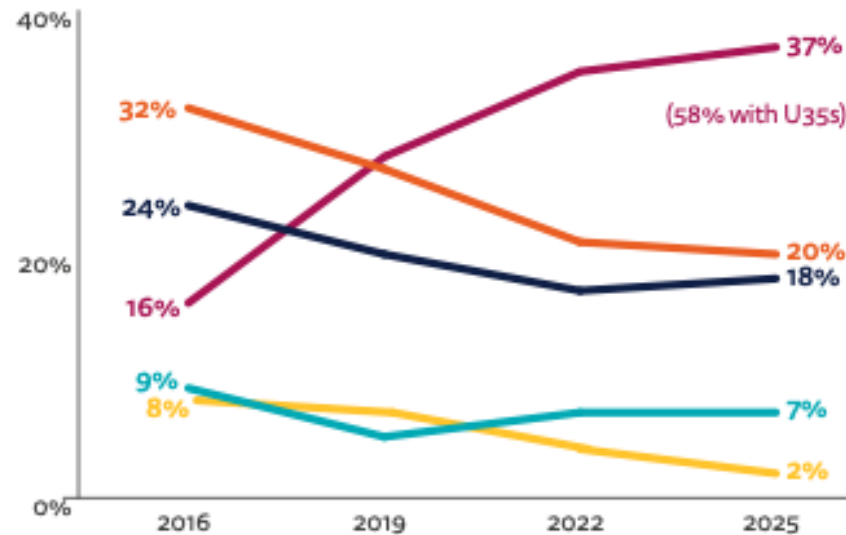
News media: what is it good for?

PROPORTION THAT SAY EACH IS THE FIRST WAY THEY COME ACROSS NEWS IN THE MORNING (2016-2025) - UK AND USA

Over the last decade the smartphone has stolen attention from other devices and sources

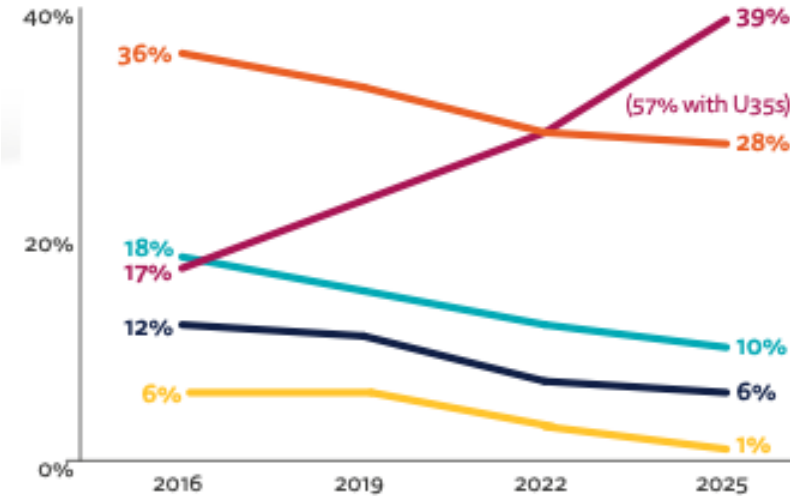
UNITED KINGDOM

TV Radio Smartphone Computer Newspaper



UNITED STATES

TV Radio Smartphone Computer Newspaper



Q9c_new2016. What is the FIRST way you typically come across news in the morning? Base: Total sample in each country-year = 2000.

News consumption: engagement time

News source		Engagement time per day
	Print paper	 25-40 minutes
	Mobile	 15 seconds – 2 minutes

(Pew Research Center, 2023)



News & Media: moving towards the future

Production/ Distribution



Consumption





Digitalization of news: consumption

Consequences for readers

- Access to an incredible number of information sources
- Information overload – how do you prioritize interests and time spent?
 - Competition for attention – also with other platforms
 - Possibility to read about the same thing only
 - Readers want diversity and objectivity
 - Also: expect personalization (like the socials..)
 - Technically: a curated news site is possible



Algorithmic sorting of media content

Sorting and prioritizing according to specifically chosen principles

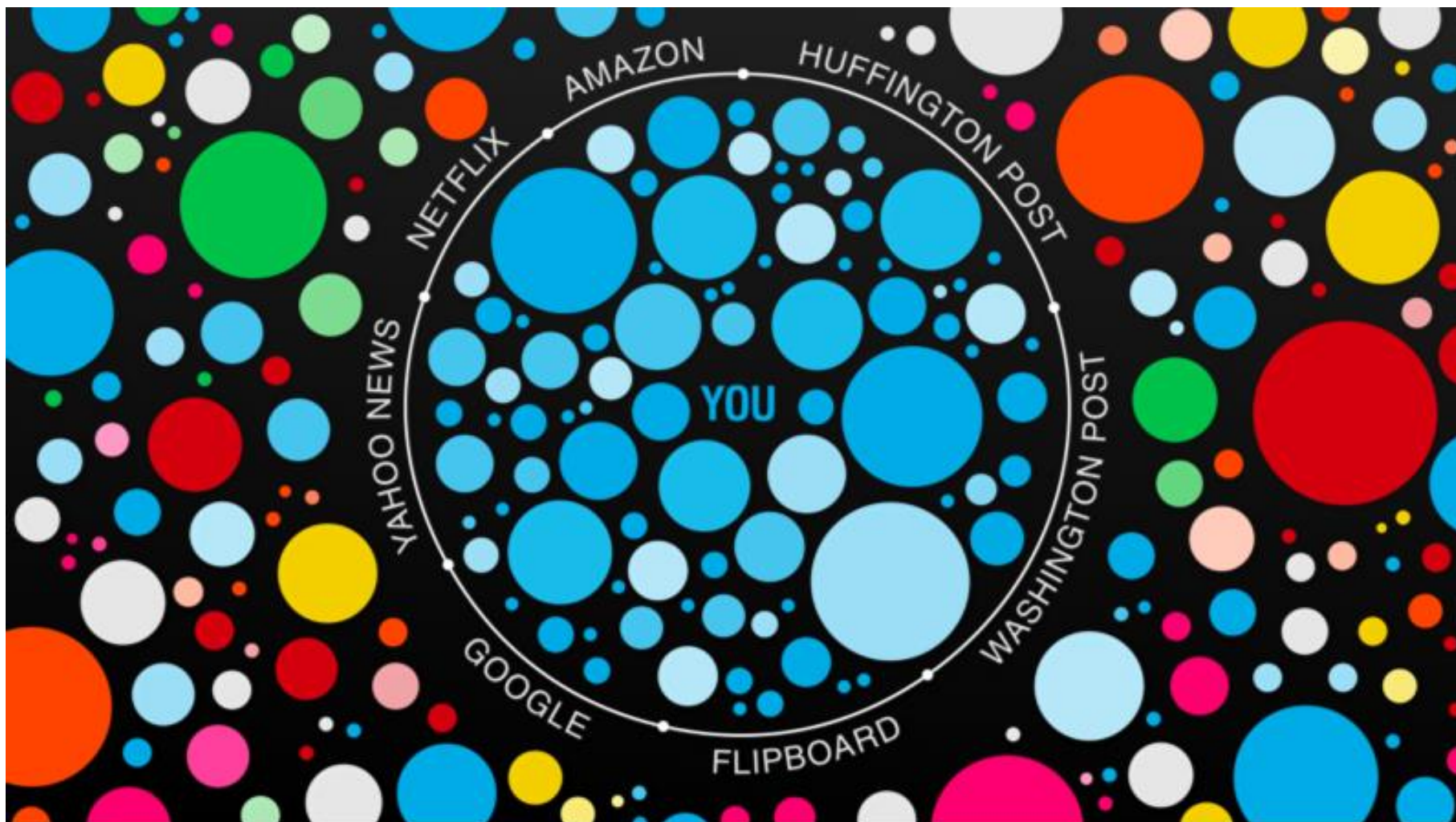
- e.g. match on key criteria with previously chosen content

Geared towards specifically chosen goals

- e.g. clicks, time spent, reading article until end, returning to website

→ Popularity, exposure bias might play a role here

→ Contribute to the threat of **filter bubbles**



Digitalization of news: consumption

Bubble trouble

- What's the problem with filter bubbles?
- How can it impact news consumption?
- What are the implications of those impacts?
- How is **algorithmic gatekeeping** different from traditional gatekeeping?



Filter bubbles and polarization



Threats: polarization, decreased diversity and the erosion of a common core



Common core... of what?

A quick (but very important!) digression



Jürgen Habermas

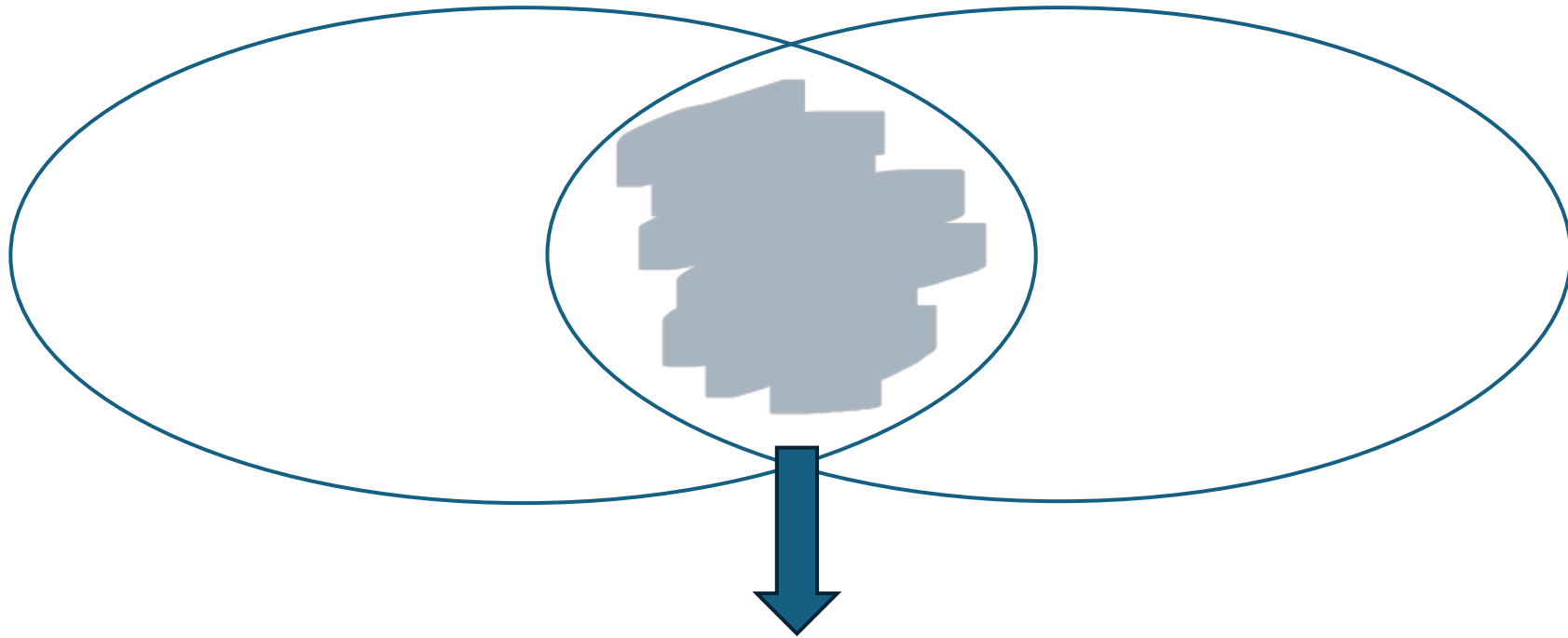
Public Sphere

A communication space where private individuals come together as a public to discuss matters of **common concern**.

Common concern: public opinion, deliberation, education, etc...

→ Deliberative Democracy

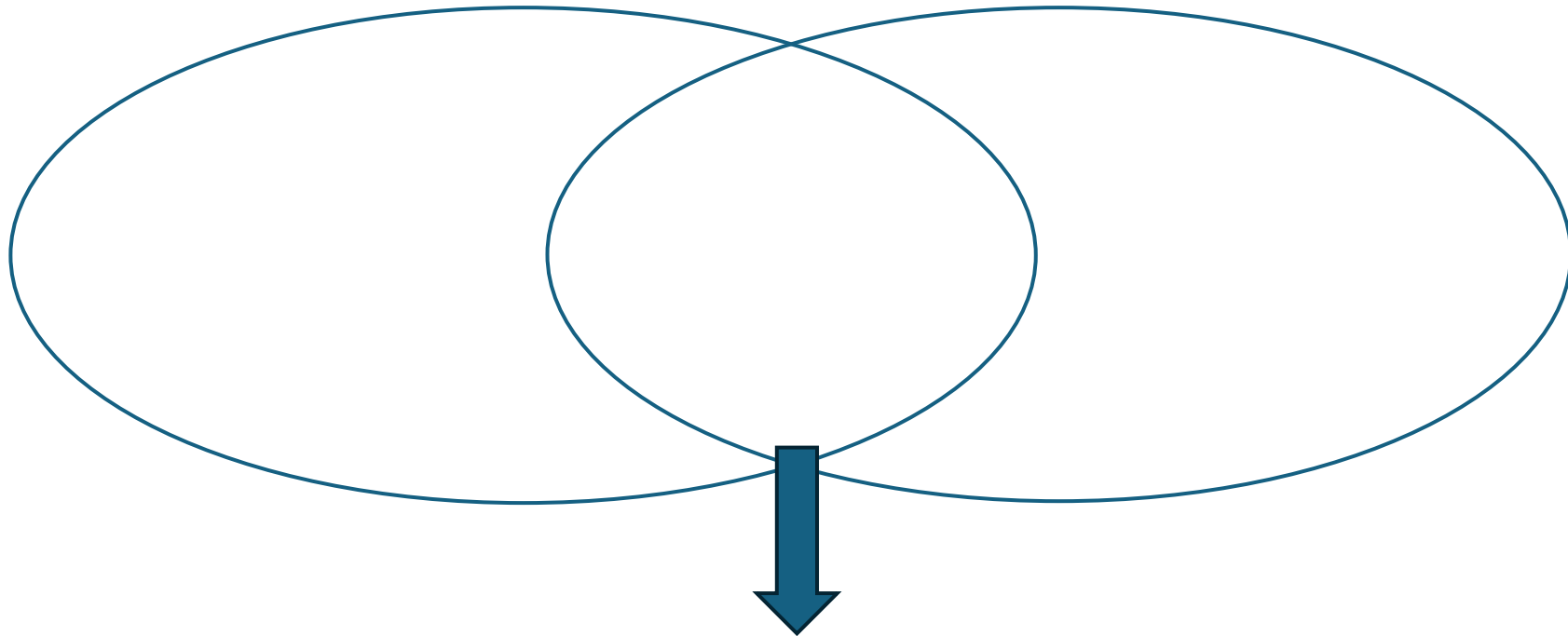
Common core... of what?



Shared topic awareness
Shared factual baselines
Mutual recognition
Cross-cutting exposure



Common core... of what?



Parallel realities
Identity signaling
Tribal reinforcement



Common core... of what?

Very important for (democratic) deliberation: **informed citizenship**

- Difference between *normative disagreement* and *epistemic fragmentation*
 - Normative disagreement: attitudes about facts or how to interpret/assign value to those facts
 - Epistemic fragmentation: arguing about what the facts are in the first place
- Debate vs. Distrust
- **How does this affect democratic decision-making?**

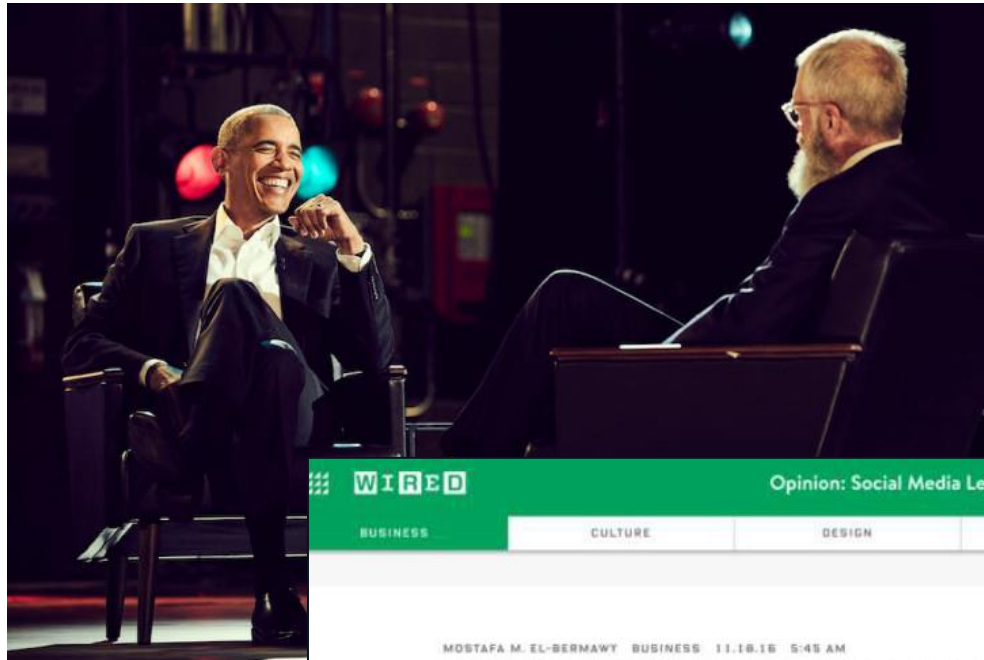
Filter bubbles and polarization



Threats: polarization, decreased diversity and the erosion of a common core



Digitalization of news: consumption







Should we worry about filter bubbles?

Does algorithmic gatekeeping decrease diversity?

Algorithmic diversity

Method: Measuring diversity of simulated news recommendation outputs based on real-world user data and recommendations by the news room (N=21,973 articles, N=500 users)

Finding: Almost all recommendation settings are on par with human editors, including the user history even increased diversity

Do not blame it on the algorithm: an empirical assessment of multiple recommender systems and their impact on content diversity

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ABSTRACT

In the debate about filter bubbles caused by algorithmic news recommendation, the conceptualization of the two core concepts in this debate, diversity and algorithms, has received little attention in social scientific research. This paper examines the effect of multiple recommender systems on different diversity dimensions. To this end, it maps different values that diversity can serve, and a respective set of criteria that characterizes a diverse information offer in this particular conception of diversity. We make use of a data set of simulated article recommendations based on actual content of one of the major Dutch broadsheet newspapers and its users ($N=21,973$ articles, $N=500$ users). We find that all of the recommendation logics under study proved to lead to a rather diverse set of recommendations that are on par with human editors and that basing recommendations on user histories can substantially increase topic diversity within a recommendation set.

ARTICLE HISTORY

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Accepted 20 February 2018

KEYWORDS

News; recommender systems; diversity metrics; filter bubbles; automated content classification

1. Introduction

Functioning democracies rely on the media's role to enable different groups and interests in society to participate in the democratic debate and create the opportunity for citizens to encounter diverse opinions (Owen & Smith, 2015). Therefore, diversity is one of the core public values in media law and policy. As the Council of Europe (2007) has put it, 'media pluralism and diversity of media content are essential for the functioning of a democratic society' and the right to freedom of expression 'will be fully satisfied only if each person is given the possibility to form his or her own opinion from diverse sources of information'. The close link between media diversity and democratic participation may also explain the vigor with which new technological innovations in the media landscape are being met. The impact of personalized recommendations and algorithmic filtering on users' information diet is subject to dystopian visions about filter bubbles and echo chambers. But how much

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Attitudes towards news curation

Method: Comparative survey in 26 countries (N = 53,314). Multi-level models to predict attitudes towards editorial, peer, and algorithmic gatekeeping

Finding: Audiences believe algorithmic selection guided by a user's past consumption behavior is a better way to get news than editorial curation.

But: variation on the individual level.

OPEN ACCESS

MY FRIENDS, EDITORS, ALGORITHMS, AND I

Examining audience attitudes to news selection

Neil Thurman, Judith Moeller, Natali Helberger and Damian Trilling

Prompted by the ongoing development of content personalization by social networks and mainstream news brands, and recent debates about balancing algorithmic and editorial selection, this study explores what audiences think about news selection mechanisms and why. Analysing data from a 26-country survey (N = 53,314), we report the extent to which audiences believe story selection by editors and story selection by algorithms are good ways to get news online and, using multi-level models, explore the relationships that exist between individuals' characteristics and those beliefs. The results show that, collectively, audiences believe algorithmic selection guided by a user's past consumption behaviour is a better way to get news than editorial curation. There are, however, significant variations in these beliefs at the individual level. Age, trust in news, concerns about privacy, mobile news access, paying for news, and six other variables had effects. Our results are partly in line with current general theory on algorithmic appreciation, but diverge in our findings on the relative appreciation of algorithms and experts, and in how the appreciation of algorithms can differ according to the data that drive them. We believe this divergence is partly due to our study's focus on news, showing algorithmic appreciation has context-specific characteristics.

KEYWORDS algorithms; collaborative filtering; gatekeeping; journalistic curation; news selection; personalization; recommender systems; user tracking

Introduction

Algorithmic recommendations have become central to how consumers interact with some media outlets. Netflix estimates that its recommender system influences choices responsible for 80 per cent of the video streamed on its platform and saves the company more than \$1 billion per year by increasing customer retention (Gomez-Urbe and Hunt 2015, 5).

While recommender systems may not become as central to the business of journalism, news publishers continue to experiment with, and extend the provision of, personalized news (see e.g. Kunert and Thurman, in press), as evidenced by recent

Social media news use and polarization?

Method: Combining web-tracking data with survey data to link preferred access mode to news with political attitudes and behavior (N=302).

Finding: No relationship between preference for curated news feeds and extreme positions on political ideology, **but** strong relationship with political interest and trust in media.



Do users understand what is happening?

Method: Focus groups among German media users to

Finding: Contradiction in how users perceive themselves as active participants in the process, but lack the means to exercise agency; news personalization strategies afford new opportunities for introducing reciprocity in the media-audience relationship, **but** it is complicated.



Should we worry about filter bubbles?

Does algorithmic gatekeeping decrease diversity?

- Algorithmic filtering can increase diversity compared to human gatekeeping (Moeller, Trilling, van Es, & Helberger, 2018)

Do users accept algorithmic gatekeeping?

- Media users by and large accept and appreciate algorithmic news curation (Thurman, Moeller, Helberger, & Trilling, 2019)

Does accessing news on social media polarize users?

- Users accessing news primarily through social media are not more polarized. (Moeller, van der Velde, Merten, & Puschmann, 2020)

How do users negotiate their own role vis-à-vis NRS?

- Users want to play an active role in the curation of their news recommendations, but it is complicated. (Monzer, Moeller, Helberger, & Eskens, 2020)

Should we worry about filter bubbles?

We also do not (yet...) live exclusively c



Should we worry about filter bubbles?

“We conclude that – in spite of the serious concerns voiced – at present, there is no empirical evidence that warrants any strong worries about filter bubbles. Nevertheless, the debate about filter bubbles is important. Personalization on news sites is still at an infant stage, and personalized content does not constitute a substantial information source for most citizens, as our review of literature on media use has shown.

(Zuiderveen Borgesius et al., 2016, p. 10)





Digitalization of news: production

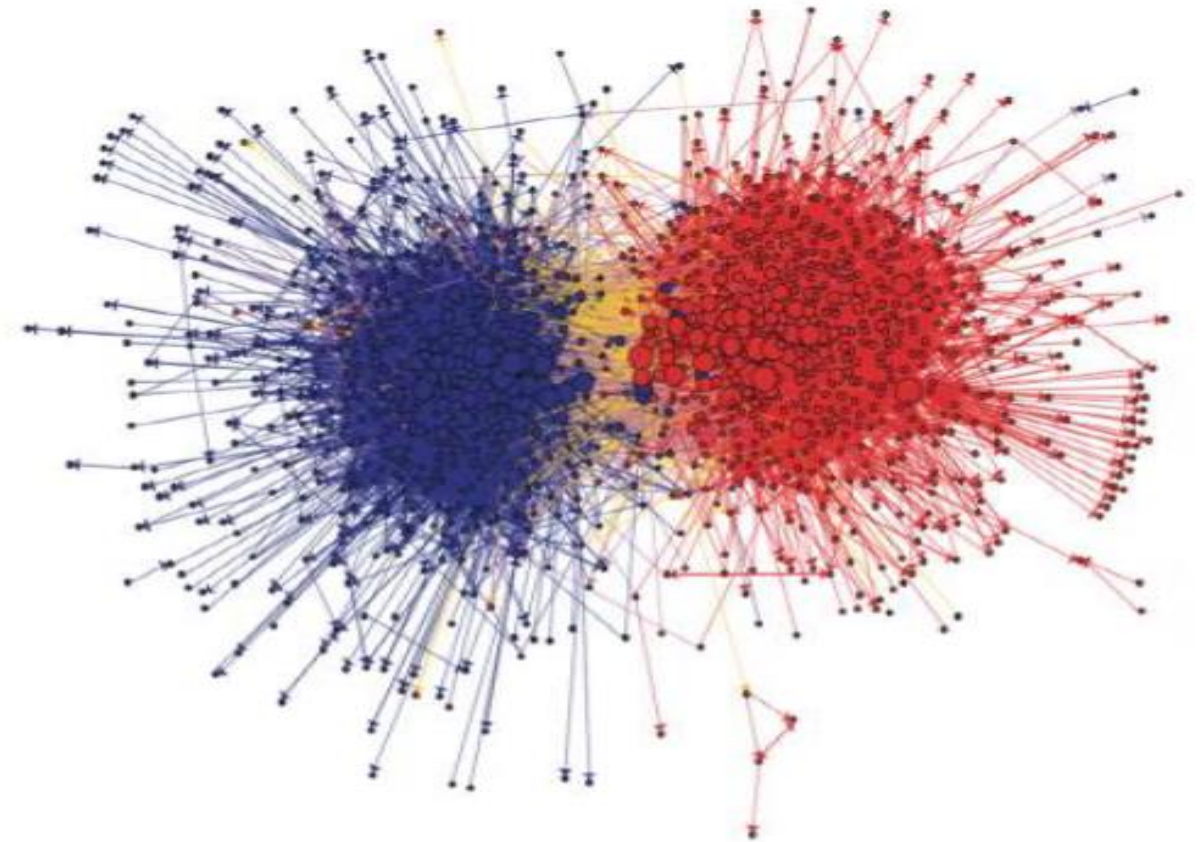
News for you!

Echo chambers vs. Filter Bubbles

Presidential blogs during the 2004 election: Liberal and conservative link to the same sources

- Internal filtering
- Little to do with technology
- Human decision making
(remember *Human vs System* bias?)

Connection vs. engagement



Adamic & Glance (2005)

News for you!

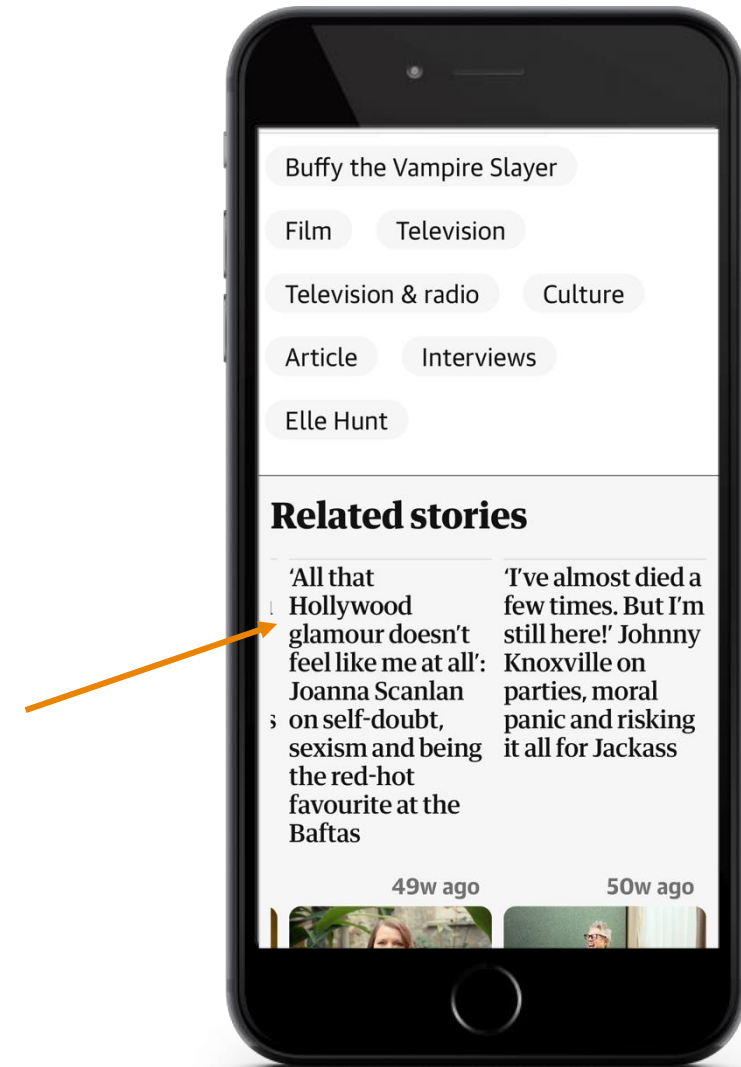
News recommender systems (NRS)

- Couldn't we just take the algorithms we know from Social Media and apply them to the news?
 - Different from recommenders on social media
 - Selecting a news website is “filtering” in and of itself
 - Once you're in the news environment, different rules, expectations, responsibilities and regulations apply
- Can you reflect on some of these?

News for you!

News recommender systems (NRS)

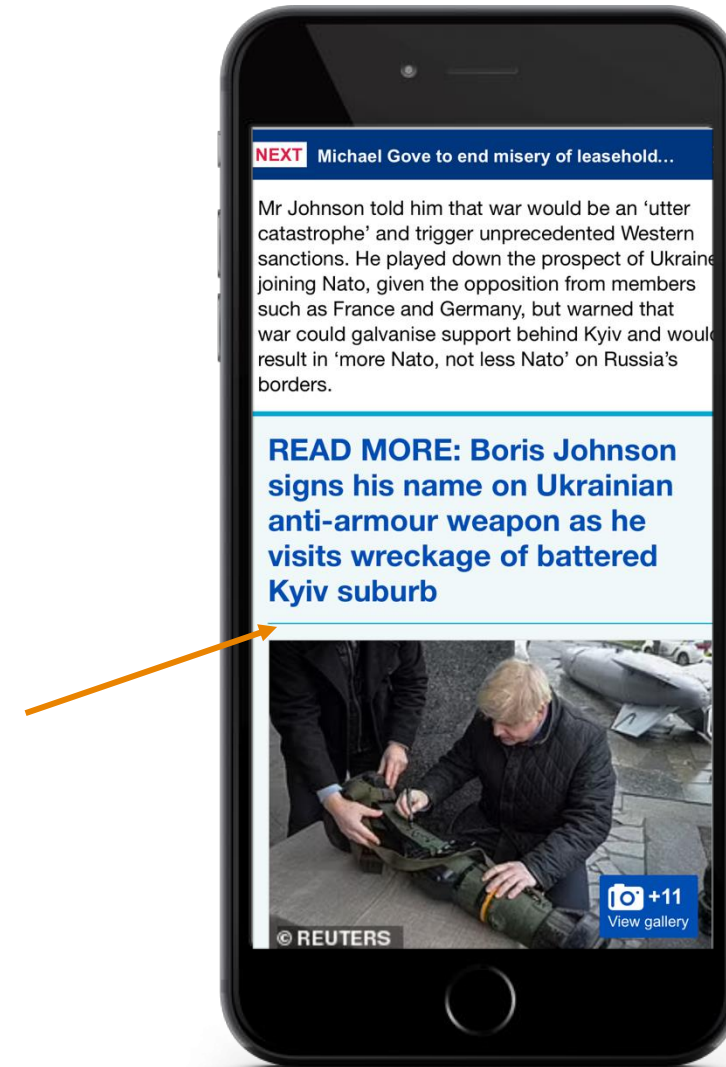
- Algorithms that can make automated and/or personalized recommendations based on metadata, past behavior, and/or ratings of similar users (Ricci et al., 2011)
- NRS can be used anywhere from recommendations below articles, embedded within content, in special sections, for newsletters, etc.



News for you!

News recommender systems (NRS)

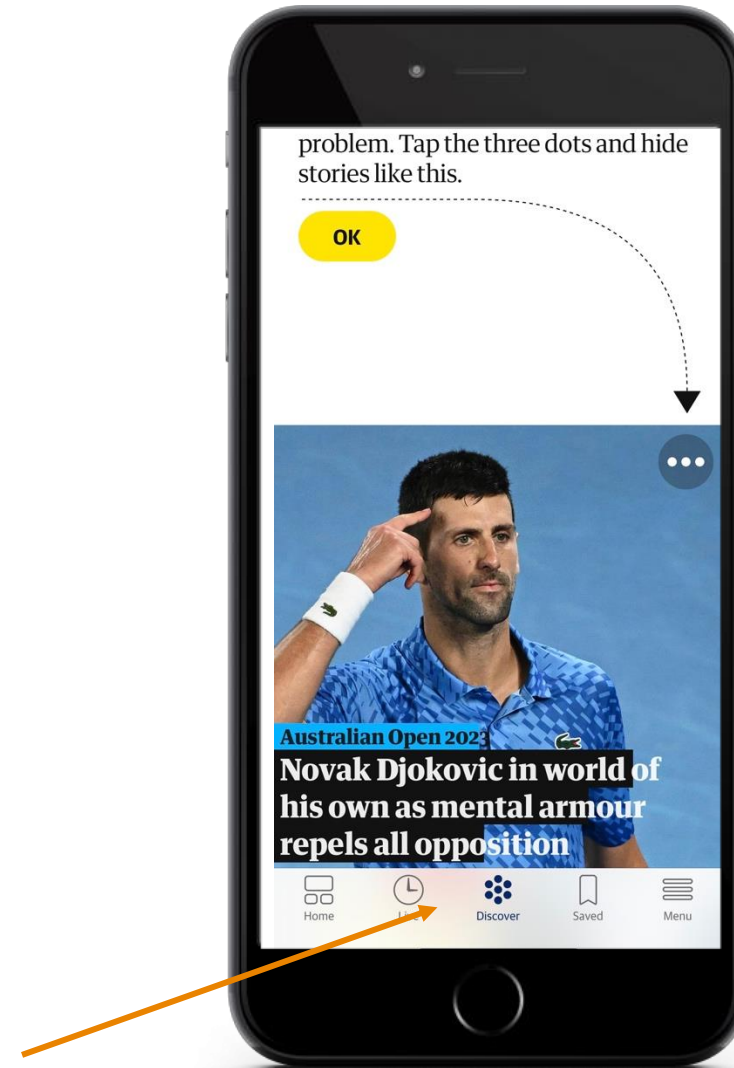
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News for you!

News recommender systems (NRS)

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Current state of NRS: the experts weigh in

Interviews with experts in Journalism revealed that...

- Most news organizations are still in a **conceptual phase**
 - Algorithmic recommenders are a topic but by far not *the* topic
- There are many **benefits** of NRS to the newsroom
 - Better value to the reader, responsiveness, exposure of archive, in-depth reading, and ultimately: more subscriptions
- Yes, there are **concerns**: Loss of editorial power, becoming too user focused
 - Mostly, there are only theoretical

Current state of NRS: the experts weigh in

Interviews with experts in Journalism revealed that...

- There is a shift within **organizational dynamics** in the newsroom
 - New crop of people (data scientists) 
 - Lack of a “common language”
- Any concerns and tensions are minimal in practice
 - Retention of **editorial control** will (at least for now) be the priority



“It is called *Artificial Intelligence*, but these things are quite stupid, actually.”

News for you!

News recommender systems (NRS)

- **Yes:** they are used in news production/distribution
- **No:** they are not as powerful/dangerous as we think they are (at least not yet...)





NRS: what does the future look like?

NRS: looking to the future

Focus on “Responsible NRS” – what does that mean?

- Overall goal: inform the public (in a way that reflects the org’s values)
- Increase visibility of content (new & old)
- Better user experience
- Cooperation between departments
- Economic success through responsiveness
- “Bilingual” experts

Production/ Distribution



NRS: looking to the future

Focus on “Responsible NRS” – what does that mean?

- Use recommenders to increase diversity
- Encourage critical thinking and engagement with content
- Lure people out of their bubble
- Facilitate exchange and discussion

Consumption





You tell me! IN GROUPS:

Focus on “Responsible NRS” – if you build it, they will come!

Build a recommender that defends democracy (actually)

- Think about:
 - Visibility of content, user experience
 - Data/Information used
 - Economic success through responsiveness
 - Diversity

Present this to the class (5 mins maximum! 1 slide minimum!) 



NRS: looking to the future

This is assuming that we make it to the news site...

NRS: looking to the future

The end of NRS??

- What if we never make it to the news site?
- Search engines → endpoint of (political/news) information search?
 - Generative AI can (soon) give us comprehensive answers to our questions.
 - What does that mean for our information environment?
 - What does that mean for opinion formation?
 - What could possibly go wrong...?

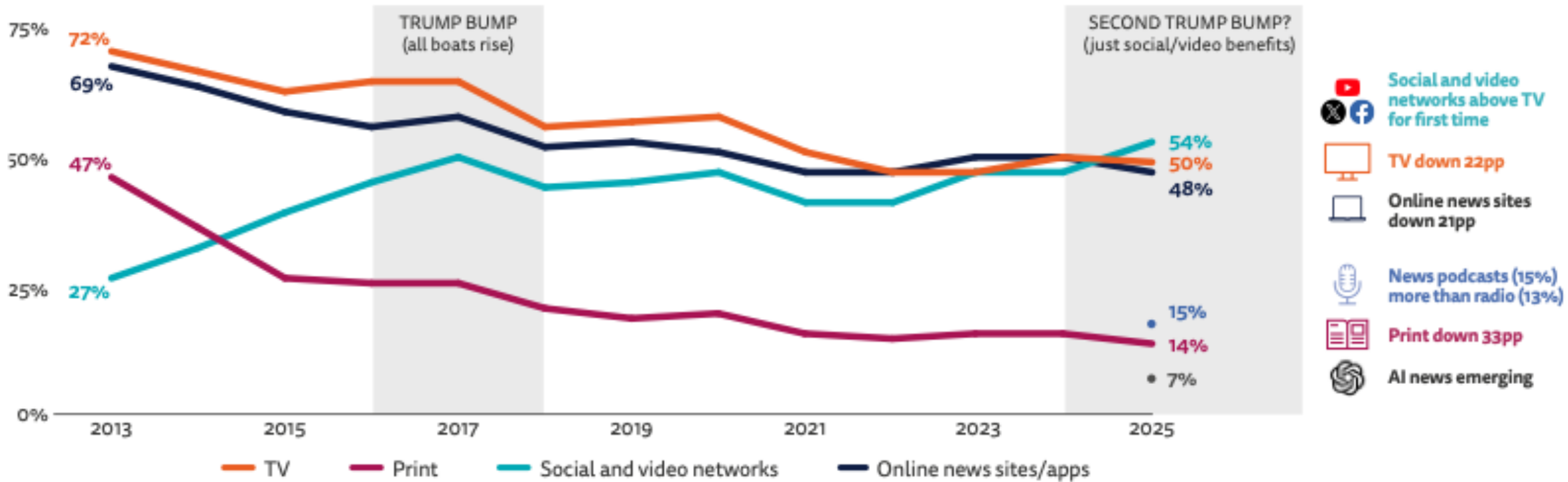
Should we worry about filter bubbles?

“We conclude that – in spite of the serious concerns voiced – at present, there is no empirical evidence that warrants any strong worries about filter bubbles. Nevertheless, the debate about filter bubbles is important. Personalization on news sites is still at an infant stage, and personalized content does not constitute a substantial information source for most citizens, as our review of literature on media use has shown. However, if personalization technology improves, and personalized news content becomes people’s main information source, **problems for our democracy could indeed arise**, as our review of empirical studies of media effects has shown.”

(Zuiderveen Borgesius et al., 2016, p. 10)

NRS: looking to the future

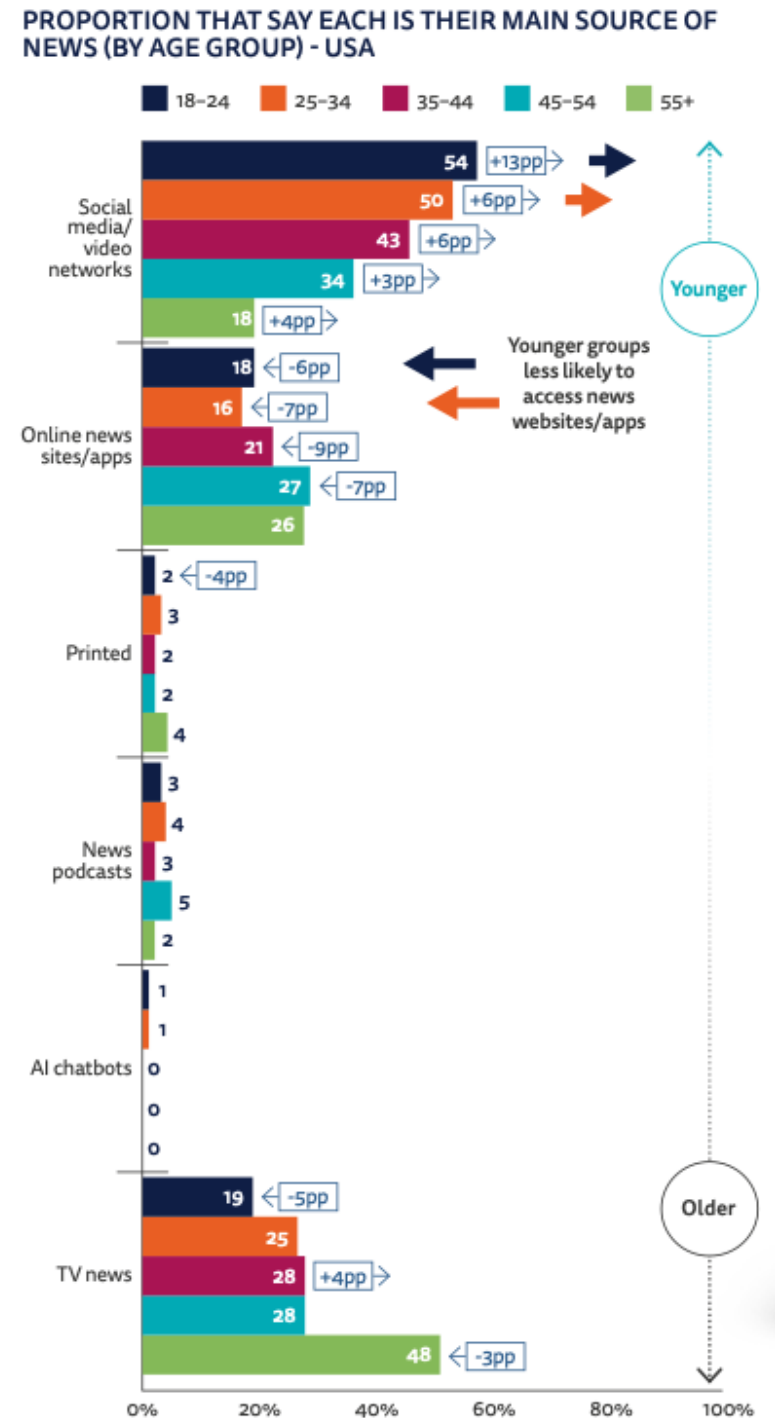
PROPORTION THAT USED EACH AS A SOURCE OF NEWS IN THE LAST WEEK (2013-2025) - USA



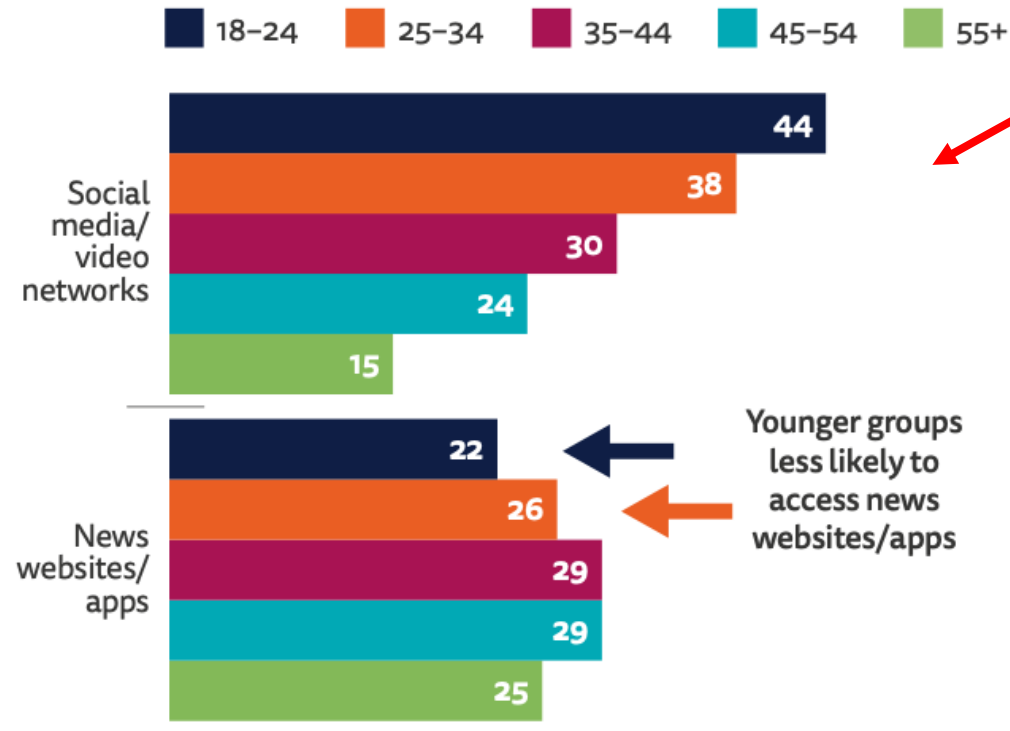


NRS: looking to the future

Digital News Report
Newman, 2025

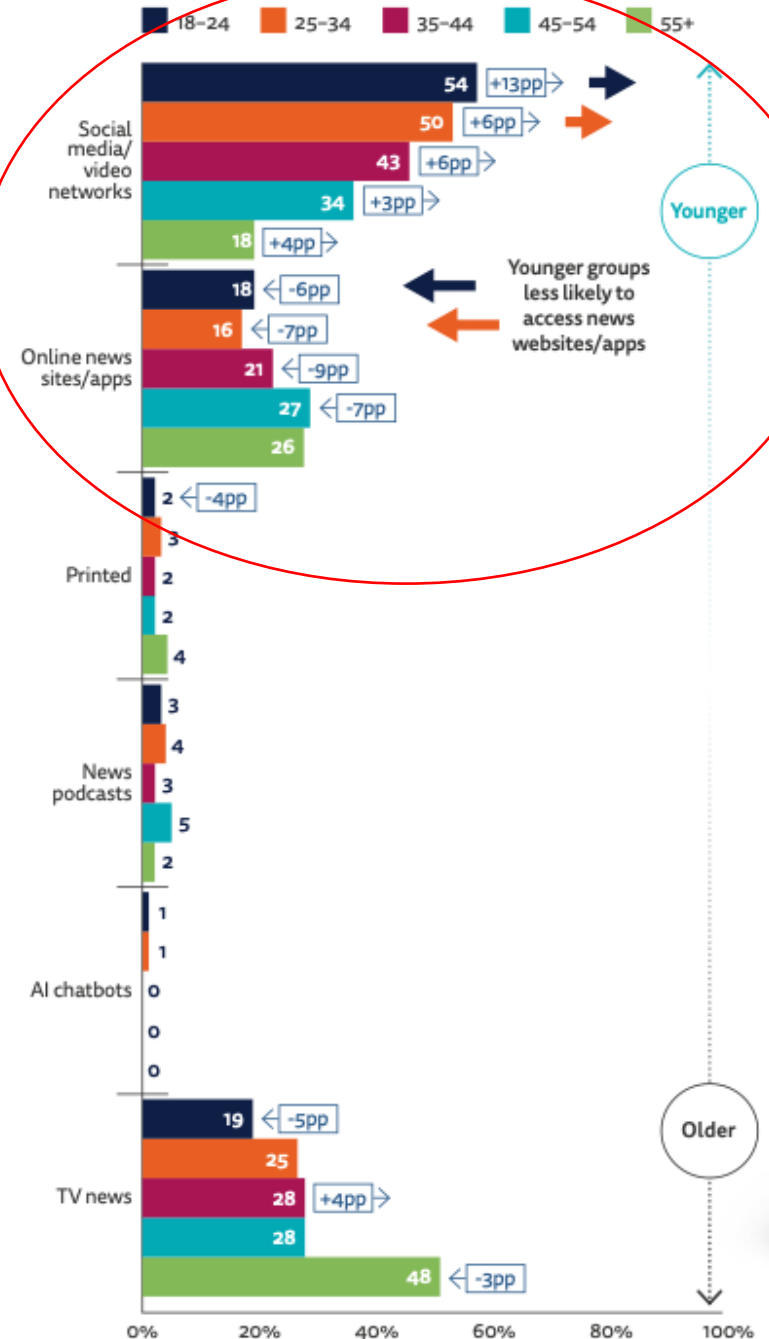


NRS: looking to the future



Digital News Report
Newman, 2025

PROPORTION THAT SAY EACH IS THEIR MAIN SOURCE OF NEWS (BY AGE GROUP) - USA



NRS: looking to the future

TOP REASONS FOR NEWS AVOIDANCE – ALL MARKETS



39%

say news has negative effect on mood



31%

say they are worn out by the amount of news



30%

say there is too much coverage of conflict/war



29%

say there is too much coverage of politics



20%

say there is nothing I can do with the information



18%

say it leads to arguments I'd rather avoid



18%

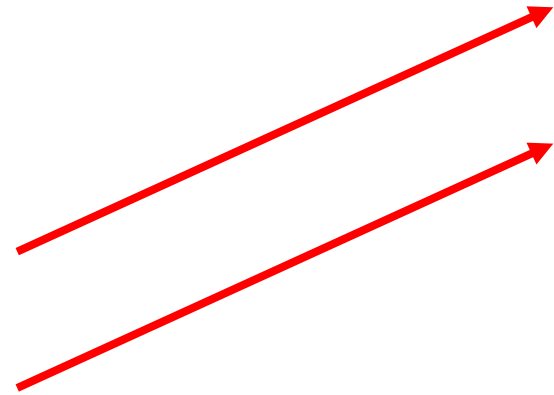
say it doesn't feel relevant to my life



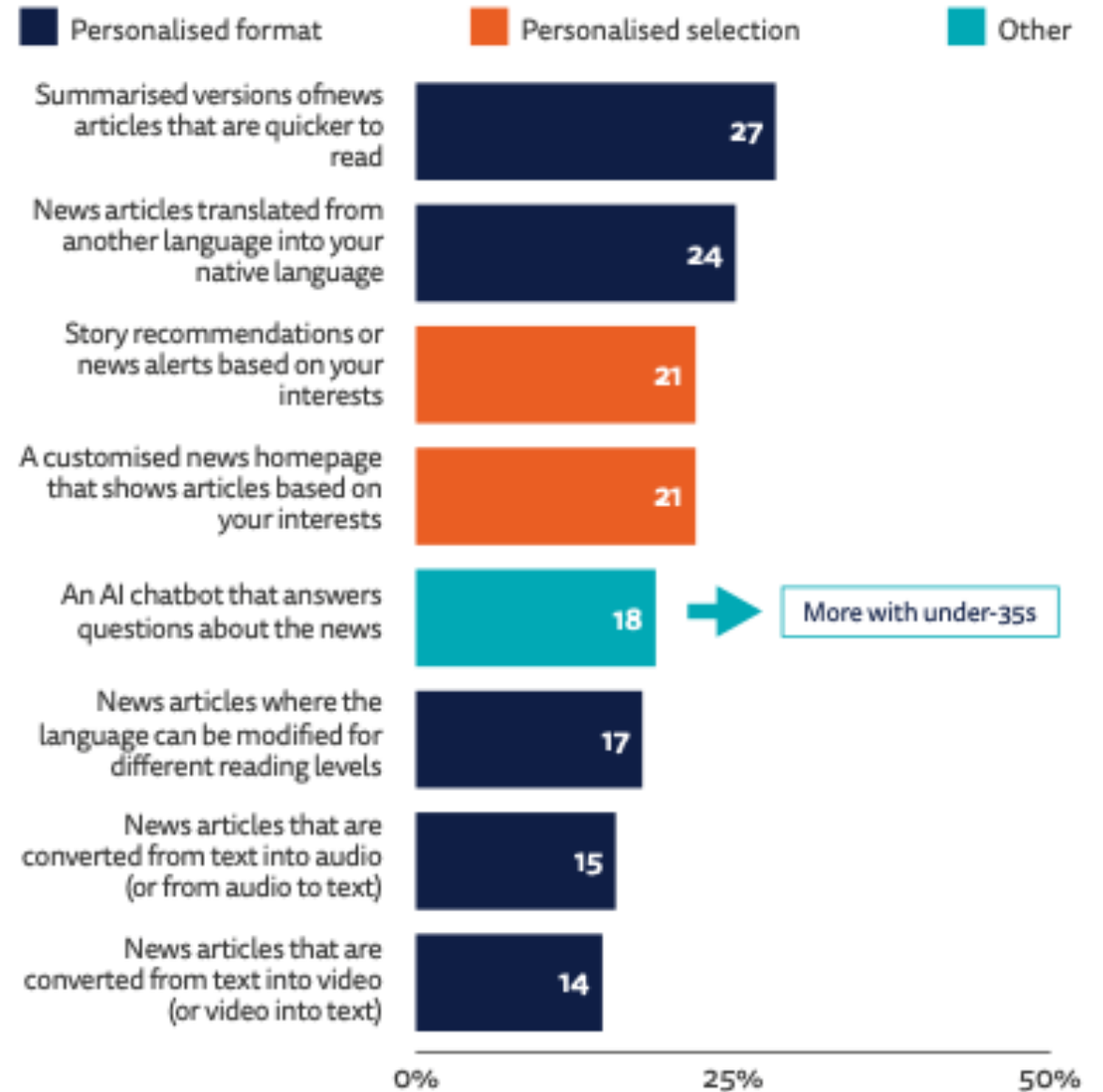
9%

say news is too hard to understand
(14% with U35s)

NRS: looking to the future

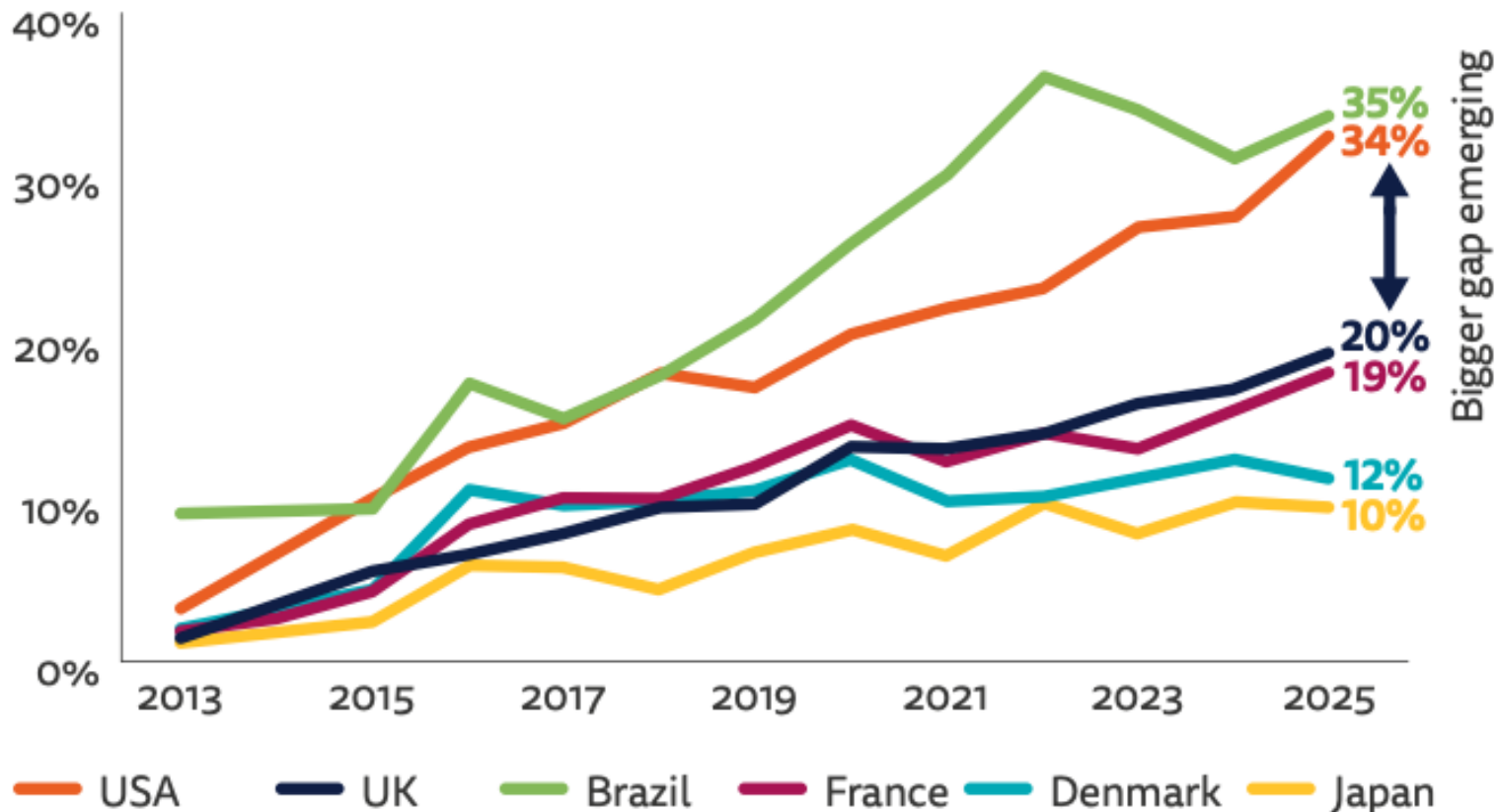


PROPORTION THAT SAY THEY WOULD BE INTERESTED IN USING EACH TYPE OF PERSONALISATION – ALL MARKETS



NRS: looking to the future

PROPORTION THAT SAY SOCIAL MEDIA IS THEIR MAIN SOURCE OF NEWS (2013-2025) - SELECTED COUNTRIES



Social media and news: same but different?

The stakes are different for news

News organizations differ from social media in terms of business models

- Social media: offers the same sort of content - and you generally like finding the content that you like. → Everyone's happy(ish)!
- News organizations: want to inform you. They would like to show you different perspectives and offer you **a greater diversity** instead of the same kind of information
 - AND they would still like to keep you interested and keep you reading → even if it is just for 15 seconds!

Social media and news: same but different?

New solutions needed to tackle this issue



Summary

- News recommender systems are in use – but this varies by platform
- The questions of democracy – what it is and how it can be deemed “successful” – plays a big part in this.
- Gatekeeping information is a crucial aspect of the political and technical discussion
- Filter bubbles are real but not as prevalent (on news sites) as we think
- Future of NRS has many riddles to solve: what are fair recommenders? And what happens when Social Media becomes the gatekeeper?